

SSN College of Engineering, Kalavakkam
Department of Computer Science and Engineering
V Semester - CSE 'A'
UCS2511 NETWORKS LAB
Exercise 13
Simulation of Link State Routing Protocol

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Aim:

Simulation of **Link State Routing** Protocol

Code:

```
set ns [new Simulator]
```

```
$ns color 1 red
```

```
$ns color 2 blue
```

```
set n0 [$ns node]
```

```
set n1 [$ns node]
```

```
set n2 [$ns node]
```

```
set n3 [$ns node]
```

```
set n4 [$ns node]
```

```
set n5 [$ns node]
```

```
set n6 [$ns node]
```

```
set n7 [$ns node]
```

```
set n8 [$ns node]
```

```
set n9 [$ns node]
```

```
set n10 [$ns node]
```

```
set n11 [$ns node]
```

```
set nr [open out.tr w]
```

```
$ns trace-all $nr
```

```
set nf [open out.nam w]
```

```
$ns namtrace-all $nf
```

```
proc finish { } {
```

```
global ns nr nf
```

```
$ns flush-trace
```

```
close $nf
```

```
close $nr
```

```
exec nam out.nam &
```

```
exit 0
}
```

```
# $ns duplex-link $n0 $n1 1Mb 10ms DropTail
# $ns duplex-link $n1 $n2 1Mb 10ms DropTail
# $ns duplex-link $n2 $n3 1Mb 10ms DropTail
# $ns duplex-link $n3 $n4 1Mb 10ms DropTail
# $ns duplex-link $n4 $n5 1Mb 10ms DropTail
# $ns duplex-link $n5 $n6 1Mb 10ms DropTail
$ns duplex-link $n6 $n7 1Mb 10ms DropTail
# $ns duplex-link $n7 $n8 1Mb 10ms DropTail
```

```
$ns duplex-link $n0 $n8 1Mb 10ms DropTail
$ns duplex-link $n1 $n10 1Mb 10ms DropTail
$ns duplex-link $n0 $n9 1Mb 10ms DropTail
$ns duplex-link $n9 $n11 1Mb 10ms DropTail
$ns duplex-link $n10 $n11 1Mb 10ms DropTail
$ns duplex-link $n11 $n5 1Mb 10ms DropTail
```

```
# $ns duplex-link-op $n0 $n1 orient right
# $ns duplex-link-op $n1 $n2 orient down
# $ns duplex-link-op $n2 $n3 orient down
# $ns duplex-link-op $n3 $n4 orient left
# $ns duplex-link-op $n4 $n5 orient left
# $ns duplex-link-op $n5 $n6 orient left
$ns duplex-link-op $n6 $n7 orient left
# $ns duplex-link-op $n7 $n8 orient up
$ns duplex-link-op $n8 $n0 orient up
$ns duplex-link-op $n5 $n11 orient up
$ns duplex-link-op $n0 $n9 orient right-down
$ns duplex-link-op $n9 $n11 orient right
$ns duplex-link-op $n11 $n10 orient right
$ns duplex-link-op $n1 $n10 orient left-down
```

```
set udp0 [new Agent/UDP]
$ns attach-agent $n0 $udp0
$udp0 set fid_ 1
```

```
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 1000
$cbr0 set rate_ 0.15mb
$cbr0 set random_ true
$cbr0 attach-agent $udp0
```

```
set null0 [new Agent/Null]
$ns attach-agent $n5 $null0
```

```
$ns connect $udp0 $null0

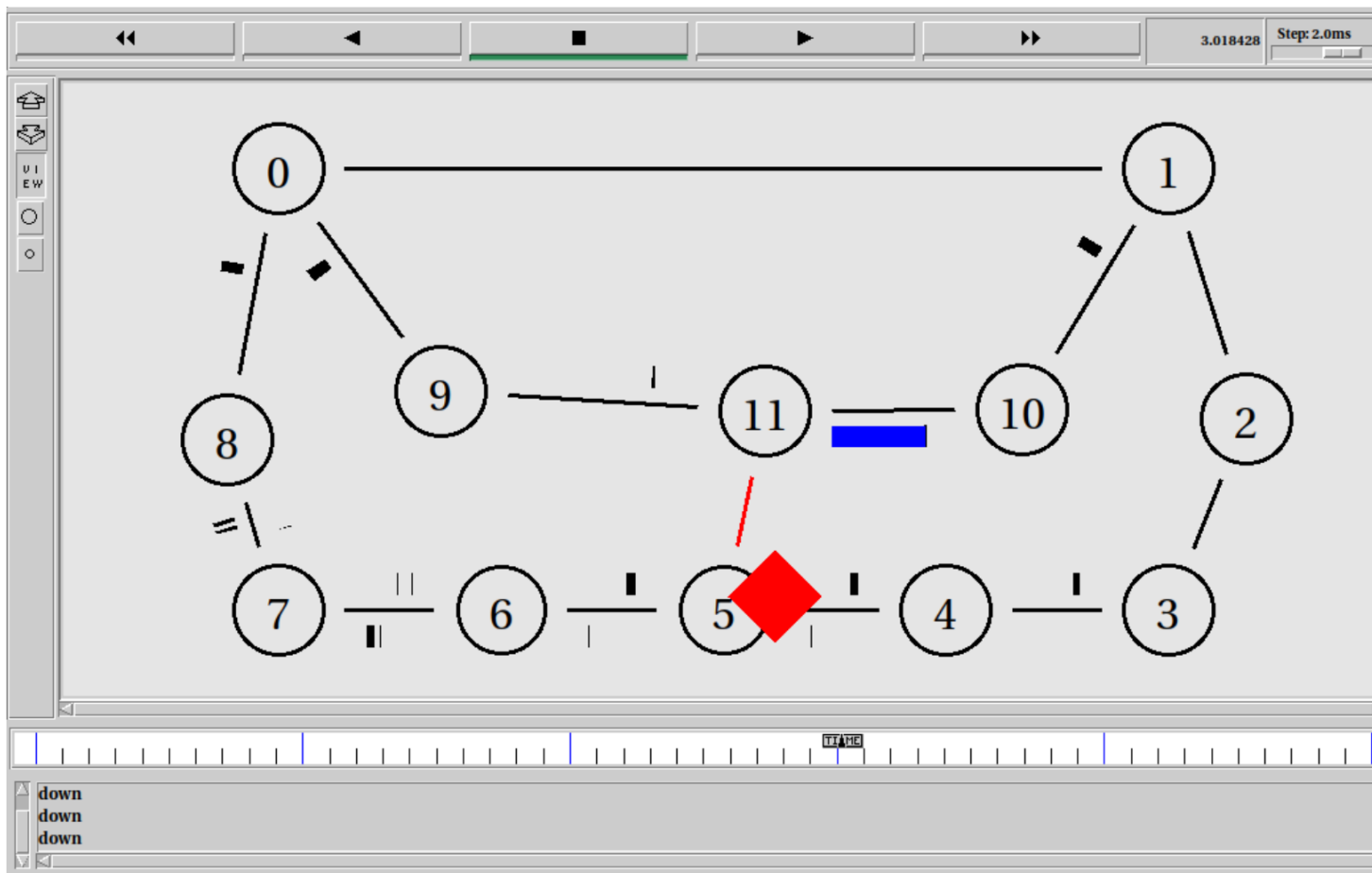
set udp1 [new Agent/UDP]
$ns attach-agent $n1 $udp1
$udp1 set fid_ 2

set cbr1 [new Application/Traffic/CBR]
$cbr1 set packetSize_ 1000
$cbr1 set rate_ 0.10mb
$cbr1 set random_ true
$cbr1 attach-agent $udp1

set null1 [new Agent/Null]
$ns attach-agent $n5 $null1
$ns connect $udp1 $null1

$ns rtproto LS
$ns rtmodel-at 3.0 down $n11 $n5
$ns rtmodel-at 2.0 down $n7 $n6
$ns rtmodel-at 4.0 up $n11 $n5
$ns rtmodel-at 3.0 up $n7 $n6
$ns at 0.0 "$cbr0 start"
$ns at 0.75 "$cbr1 start"
$ns at 5.0 "finish"
$ns run
```

Output:



Learning Outcomes:

I learned how to implement Link State Routing protocol.